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SCHOOL OF MATHEMATICS AND NATURAL SCIENCES

A PROJECT REPORT

ON

"DISTRIBUTED DATA PROCESSING PIPELINE"

Submitted

By

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Under the

guidance of

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Towards

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DECLARATION

I, **Umesh Naik,** hereby declare that this project work entitled **Distributed Data Processing Pipeline** is submitted in partial fulfilment for the award of the degree of **M.Sc. Data Science** of **Chanakya University**.

I further declare that I have not submitted this project report either in part or in full to any other university for the award of any degree.

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ABSTRACT

The rapid growth of digital platforms has resulted in the generation of massive volumes of transactional data. Processing such data using traditional single-machine systems poses significant challenges in terms of scalability, performance, and reliability. Distributed data processing pipelines address these challenges by enabling parallel computation, fault tolerance, and efficient data handling across multiple nodes.

This mini project presents a conceptual and practical study of a distributed data processing pipeline applied to transaction data analytics. The project focuses on understanding the theoretical foundations of distributed pipelines, including data ingestion, distributed storage, parallel processing, and fault tolerance mechanisms. A transaction dataset containing user purchases across multiple categories and timestamps is used as a case study. The practical implementation is carried out using a Jupyter Notebook to demonstrate how distributed processing concepts can be applied to real-world data.

The outcomes of the project highlight the importance of distributed data pipelines in modern data engineering and analytics workflows, particularly for large-scale data-driven applications.

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INTRODUCTION

1.1 INTRODUCTION

In recent years, the volume, velocity, and variety of data generated by digital systems have increased exponentially. Applications such as e-commerce platforms, financial systems, and online services continuously generate large-scale transactional data. Analyzing such data is crucial for gaining insights into user behavior, business trends, and operational efficiency.

Traditional data processing systems that operate on a single machine are often insufficient to handle large datasets due to limitations in memory, processing power, and fault tolerance. Distributed data processing pipelines provide a scalable solution by distributing data and computation across multiple machines. These pipelines enable efficient data ingestion, processing, and analysis while ensuring reliability and performance.

This project explores the concept of distributed data processing pipelines and demonstrates their application using a transaction dataset. The focus is on understanding the architecture and theoretical principles behind distributed pipelines before applying them in practice.

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PROBLEM STATEMENT

2.1 Problem Statement

Organizations today collect vast amounts of transactional data from multiple users and platforms. Processing and analyzing this data using traditional centralized systems leads to challenges such as high latency, limited scalability, and increased risk of system failure.

The problem addressed in this project is to understand how distributed data processing pipelines can be used to efficiently handle large transaction datasets. The objective is to design and analyze a pipeline that supports scalable data ingestion, parallel processing, and reliable analytics, while extracting meaningful insights from transaction data.

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METHODOLOGY

3.0 Data Overview

The dataset used in this project consists of transaction-level records representing user purchases. Each record includes the following attributes:

- **transaction_id:** Unique identifier for each transaction
- **user_id:** Identifier for the user performing the transaction
- **category:** Product category such as Home, Electronics, Clothing, or Sports
- **amount:** Transaction amount
- **timestamp:** Date and time of the transaction

A sample of the dataset is shown below:

transaction_id	user_id	category	amount	timestamp
cd8dc02f…	525	Home	199.52	2024-11-22
e087222a…	196	Home	83.56	2024-08-16
c53dbcb7…	40	Electronics	1692.04	2024-02-22

3.1 Data Ingestion

The first stage of the methodology involves ingesting transaction data from CSV files into the processing environment. The data ingestion process ensures that data is read in a structured format with appropriate schema definition. This step is crucial for maintaining data consistency and accuracy throughout the pipeline.

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3.2 Distributed Storage

After ingestion, the data is conceptually stored in a distributed storage system. Distributed storage allows large datasets to be partitioned and replicated across multiple nodes. This improves fault tolerance and ensures data availability even in the event of system failures.

3.3 Parallel Data Processing

Parallel processing forms the core of the distributed data pipeline. The dataset is divided into partitions, and each partition is processed independently across multiple computing nodes. Operations such as aggregation, filtering, and transformation are performed in parallel, significantly reducing execution time compared to sequential processing.

3.4 Analytical Computation

The processed data is used to compute key analytical metrics such as category-wise spending and time-based transaction trends. These metrics provide meaningful insights into user behavior and spending patterns.

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CSV Data Source
↓
Data Ingestion Layer
↓
Distributed Storage
↓
Parallel Processing Engine
↓
Analytics & Insights

Figure 3.1: Distributed Data Processing Pipeline Architecture

3.5 Key metrics and KPIs

The following metrics are used to analyze the transaction data:

- **Total Spend per Category:** Sum of transaction amounts for each category
- **Transaction Trend Over Time:** Aggregated spending across time periods
- **User-Level Spending Patterns:** Distribution of spending across users

These metrics help in understanding customer behavior and identifying trends within the dataset.

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INSIGHTS AND ANALYSIS

4.1 INSIGHTS

Based on the analysis performed using the distributed data processing approach, the following insights were observed:

- Electronics category contributes the highest overall transaction value.
- Spending patterns vary significantly across different time periods.
- Certain users exhibit higher transaction frequency and spending behavior.

The distributed processing pipeline enables efficient aggregation and analysis of these metrics, even when dealing with large datasets. The implementation demonstrates how parallel processing improves performance and scalability.

Analysis Result	Derived Insight
Electronics category shows highest total transaction amount	Customers spend more on high-value electronic products compared to other categories
Home and Clothing categories show moderate spending	These categories represent frequent but lower-value transactions
Monthly transaction values fluctuate across time periods	User spending behavior varies based on seasonality and time
Certain users contribute higher cumulative spending	A small group of users accounts for a significant portion of total revenue
Aggregation operations executed efficiently	Distributed processing enables faster computation on large datasets

CONCLUSION

This mini project presented a study of distributed data processing pipelines applied to transaction data analytics. The project emphasized the theoretical foundations of distributed pipelines, including scalability, parallel processing, and fault tolerance. Using a transaction dataset, the project demonstrated how these concepts can be applied to derive meaningful insights.

The results highlight that distributed data processing pipelines are essential for handling large-scale data in modern applications. The project provides a strong foundation for further exploration into advanced big data technologies and real-time analytics systems.

Dataset: https://www.kaggle.com/datasets/umeshnaik01/sales-transactions.

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